Concept Paper

Evaluating and Detecting Hallucinations Across Large Language Models

Aaditya Dalal
U.I.E.T, Maharshi Dayanand University, Rohtak, Haryana

PROBLEM DESCRIPTION

Large Language Models (LLMs) are increasingly deployed in high-stakes domains such as healthcare, legal research, and education. Despite their fluency, they frequently generate factually incorrect or unsupported content, a phenomenon known as **hallucination**. Automated hallucination detection is a critical open problem in AI reliability and safety.

This project frames hallucination detection as a **binary classification task**: given an LLM-generated response to a factual question, predict whether it contains a hallucination. A secondary objective is to **compare hallucination rates across three LLMs** Gemini Flash, Llama 3.3 70B (via Groq), and Mistral 7B (via HuggingFace) surfacing behavioral differences across model families.

DATASET

Primary – HaluEval (Zhang et al., 2023): A benchmark of $\sim$10,000 question-answer pairs with binary hallucination labels, covering general knowledge, summarization, and dialogue domains. Both ground truth and hallucinated answers are provided, enabling supervised classification.

Supplementary – TruthfulQA (Lin et al., 2022): 817 questions designed to elicit hallucinations from LLMs. Each of the three target models will be queried via their respective free APIs; responses will be scored against ground truth using NLI entailment scores and BERTScore to produce a cross-model evaluation dataset.

Preprocessing will include text cleaning, embedding generation (sentence-transformers), NLI entailment score computation (DeBERTa-NLI via HuggingFace), and feature engineering covering response length, confidence markers, and semantic similarity to ground truth. Class imbalance will be addressed using SMOTE or inverse class weighting.

LITERATURE REVIEW

- **HaluEval (Zhang et al., 2023):** Introduced the HaluEval benchmark and demonstrated that even GPT-4 hallucinates significantly on knowledge-intensive tasks.
- **TruthfulQA (Lin et al., 2022):** Showed that larger models are not necessarily more truthful; GPT-3 achieved only $\sim$58% truthfulness on adversarially designed questions.
- **SelfCheckGPT (Manakul et al., 2023):** A sampling-based detection approach that checks consistency across multiple LLM outputs without requiring external knowledge, at the cost of multiple API calls per query.
- **FActScoring (Min et al., 2023):** Decomposes LLM responses into atomic claims and verifies each independently against a reference knowledge base.

This project contributes a systematic **multi-classifier benchmarking study** across three diverse LLM families, with emphasis on feature importance and cross-model failure mode analysis, an angle underexplored in the above works. Results will be discussed in the context of prior literature rather than reproduced implementations.

ALGORITHMS

Six classifiers will be trained on features extracted from LLM responses:

1. **Logistic Regression** – interpretable baseline well-suited for NLI score and embedding features
2. **Support Vector Machine (SVM)** – effective in high-dimensional text feature spaces
3. **Random Forest** – captures non-linear feature interactions; provides native feature importance
4. **Gradient Boosting (XGBoost / LightGBM)** – expected to be the strongest performing tabular classifier
5. **DeBERTa-NLI** – transformer-based end-to-end classifier via HuggingFace inference API (CPU-compatible)
6. **k-Nearest Neighbors (KNN)** – embedding-space similarity baseline

EVALUATION

Models will be evaluated using **macro F1, AUC-ROC, Precision, and Recall**, with F1 prioritized due to expected class imbalance. Validation will use stratified 5-fold cross-validation with a held-out test set for final evaluation. Hyperparameter tuning will be performed via GridSearchCV for Logistic Regression and SVM, and RandomizedSearchCV for XGBoost. Results will be contextualized against reported findings in Zhang et al. (2023) and Lin et al. (2022).

EXPECTED RESULTS

Gradient Boosting and DeBERTa-NLI are expected to achieve $>75\%$ macro F1, outperforming classical baselines. Cross-model analysis is expected to reveal that smaller open-source models hallucinate more frequently than larger proprietary ones. NLI entailment score and semantic similarity to ground truth are anticipated to be the most predictive features. Should TruthfulQA-based generation yield insufficient data, HaluEval alone fully supports the classification pipeline, with cross-model comparison as a secondary contribution. The pipeline is designed to be extensible to additional models (e.g., GPT-4o, Claude Sonnet) as future work.

LIBRARIES AND TOOLS

Library / Tool	Purpose
`scikit-learn`	Classifiers, cross-validation, GridSearchCV
`xgboost` / `lightgbm`	Gradient boosting models
`transformers` (HuggingFace)	DeBERTa-NLI inference and classification
`sentence-transformers`	Semantic embedding generation
`pandas` / `numpy`	Data manipulation and feature engineering
`matplotlib` / `seaborn`	ROC curves, confusion matrices, feature importance plots
`google-generativeai`	Gemini Flash API (free tier, no credit card required)
`groq`	Llama 3.3 70B API (free tier)
`huggingface_hub`	Mistral 7B free serverless inference endpoint
`imbalanced-learn`	SMOTE for handling class imbalance